University of Michigan

NERS 515

Nuclear Measurements Laboratory

Laboratory 3: Counting Statistics

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Abstract

This experiment examines the statistical behavior of radiation counting using a Geiger-Mueller (G-M) tube and a thorium source. We recorded 50 repeated counts with a mean near 30 counts with an interval of 0.53 seconds and 1000 repeated counts with a mean near 7 counts with an interval of 0.12 seconds. We compared the distributions to Poisson and Gaussian models. The experimental means were 30.64 and 6.87 counts, and standard deviations were 5.72 and 2.61 counts, respectively. The sample standard deviations matching the Poisson prediction $\sigma = \sqrt{\bar{x}}$ [1]. The Poisson chi-squared test gave $\chi^2 = 52.40$ for 49 degrees of freedom ($p = 0.344$) and $\chi^2 = 992.47$ for 999 degrees of freedom ($p = 0.552$), both within the normal range given by Knoll [1]. The Gaussian and Poisson models were near indistinguishable at a mean near 30, while the Poisson distribution described the asymmetric data at a mean near 7 better. These results show that the counting system produces Poisson-distributed counts.

1 Introduction

Every measurement in radiation detection is a count of random events, so repeated counts over equal time intervals fluctuate even when the source, detector, and geometry do not change. The size of this fluctuation determines the accuracy of the measurement and the precision of the system used. This precision must be understood well before meaningful uncertainties can be assigned to later results [1]. Counting statistics are used to understand the precision and uncertainties of the system of the experiment in its entirety. Counts from a working system follow a statistical model for random events, while abnormally large or small fluctuations point to a malfunction in the counting system [1].

This experiment introduces us to the statistical models used for counting data. We recorded repeated counts from the decay of a thorium source using a G-M detector. The detector was connected to a system that included an amplifier, single-channel analyzer (SCA), and counter/timer [2] at two mean values, ~ 30 and ~ 7. We compare the sample standard deviation to the predicted value to test how closely each data set follows a Poisson distribution with the chi-squared test, and compare how well Gaussian and Poisson models describe the data [2].

2 Theory

2.1 Sample Statistics

For N repeated measurements x_i, the experimental mean and sample variance are

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i, \qquad s^2 = \frac{1}{N-1}\sum_{i=1}^{N} (x_i - \bar{x})^2 , \tag{1}$$

respectively, where s is the sample standard deviation [1]. The sum is divided by $N-1$ instead of N because $\bar{x}$ was calculated from the same data (Knoll, Appendix B) [1]. The standard deviation of the mean of N independent counts is $\sigma_{\bar{x}} = \sqrt{\bar{x}/N}$ (Knoll, Eq. 3.44) [1].

2.2 Binomial, Poisson, and Gaussian Distributions

If a trial has probability p of success, the probability of exactly x successes in n independent trials is given by the binomial distribution,

$$P(x) = \frac{n!}{x!\,(n-x)!}\, p^x (1-p)^{n-x}, \tag{2}$$

which has mean $\bar{x} = np$ and variance $\sigma^2 = np(1-p)$ [1]. It follows from the fact that any one specific sequence of x successes and $n-x$ failures has probability $p^x(1-p)^{n-x}$, and that there are $n!/[x!(n-x)!]$ orderings of that sequence. In a counting experiment, n is the number of radioactive nuclei in the source and p is the probability that any one of them produces a recorded count during the counting interval. The binomial distribution therefore describes a counting measurement exactly, but n is far too large to use it directly.

When n is very large and $p \ll 1$, the binomial distribution simplifies to the Poisson distribution [1],

$$P(x) = \frac{\bar{x}^x\, e^{-\bar{x}}}{x!}. \tag{3}$$

For a constant and small probability of success, the binomial distribution reduces to $(pn)^x e^{-pn}/x!$ (Knoll, Eq. 3.23), and since $pn = \bar{x}$ this gives Eq. (3) (Knoll, Eq. 3.24). Its predicted variance equals its mean, $\sigma^2 = \bar{x}$ (Knoll, Eq. 3.28) [1]. Therefore, the standard deviation of a raw count is

$$\sigma = \sqrt{\bar{x}}. \tag{4}$$

If the mean is also large (about 25 to 30), the Poisson distribution simplifies further to the Gaussian distribution,

$$P(x) = \frac{1}{\sqrt{2\pi\bar{x}}} \exp\left[-\frac{(x-\bar{x})^2}{2\bar{x}}\right], \tag{5}$$

which is symmetric and also true for $\sigma^2 = \bar{x}$ as well [1]. When the standard deviation is not tied to the mean, the Gaussian takes the form

$$G(x) = \left(\sigma\sqrt{2\pi}\right)^{-1} \exp\left[-(x-\bar{x})^2/2\sigma^2\right] \tag{6}$$

[1]. Knoll uses the Poisson distribution for counts less than 30 and the Gaussian for counts that are greater [1]. We applied both models to both data sets, as the manual specifies [2].

2.3 Chi-Squared Test

The chi-squared parameter is defined as

$$\chi^2 = \frac{1}{\bar{x}} \sum_{i=1}^{N} (x_i - \bar{x})^2 = \frac{(N-1)\, s^2}{\bar{x}} \tag{7}$$

[1]. For a Poisson distribution, we know that $\sigma^2 = \bar{x}$, so if the data are a true Poisson sample, then $s^2 \approx \bar{x}$ and $\chi^2 \approx N - 1$. The number of degrees of freedom is $\nu = N - 1$ because $\bar{x}$ was calculated from the same data. The deviation of χ^2 from ν therefore measures how far the observed fluctuation is from that of a Poisson distribution [1].

The result is expressed as p, the probability that a random sample from a true Poisson distribution would have a larger χ^2 than measured. A perfect fit for a large sample gives $p = 0.5$ [1]. Probabilities below about 0.02 indicate abnormally large fluctuations, and probabilities above 0.98 indicate abnormally small fluctuations [1].

2.4 Error Propagation

For a derived quantity $u(x, y, z, \ldots)$ of independent variables, the standard deviation is

$$\sigma_u = \sqrt{\left(\frac{\partial u}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial u}{\partial y}\right)^2 \sigma_y^2 + \cdots} \tag{8}$$

[1].

3 Equipment and Procedure

3.1 Equipment

Details of all equipment used in this lab are listed in Table 1. The list does not include the serial numbers of the SCA and the High Voltage Power Supply because we were advised not to remove the instrument from the NIM, in case it caused other issues. In addition, we used a preamplifier, a G-M tube, and a thorium source as specified in the lab manual [2].

Purpose	**Manufacturer**	**Model**	**Serial Number**
Oscilloscope	Agilent Technologies	DSO6052A	MY44007040
High Voltage Power Supply	Ortec	556	-
Amplifier	Ortec	572A	121
Single-Channel Analyzer	Ortec	551	-
Counter/Timer	Ortec	994	2108

Table 1: List of equipment used in Laboratory 3.

3.2 Experimental Procedure

We connected the G-M tube through the preamplifier to the high voltage power supply, connecting the preamplifier output to the amplifier, then to the SCA, and then to the counter/timer. The amplifier and SCA outputs were also sent to the oscilloscope, shown in Figure 1. We increased the bias on the G-M tube to +800 V, the voltage labeled on the detector. We set the amplifier to negative polarity because the G-M tube produces negative pulses, and confirmed on the oscilloscope that the amplifier output showed pulses (Figure 2). We set the SCA's lower level discriminator (LLD) setting to be 1.6 V, where it discriminated all pulses [2].

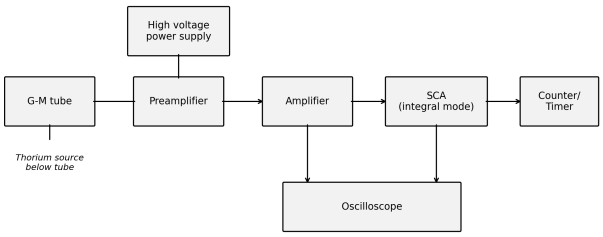


Figure 1: Block diagram of the equipment setup. The amplifier and SCA outputs were each sent to the oscilloscope.

To measure the Gaussian distribution, we needed an average amount of about 30 counts per interval. To avoid a high count rate increasing the detector's dead time, we adjusted the setup for a longer counting interval, leaving the source about as far from the detector as possible (8 cm) and adjusted only the counting interval. An interval of 0.53 s gave an average of near 30 counts, and we recorded 50 consecutive counts [2].

For the Poisson distribution, we needed an average amount of about 7 counts per interval. We kept the source 8 cm from the detector, shortened the counting interval to 0.12 s, and then recorded 1000 consecutive counts [2].

All counts were stored in a spreadsheet and analyzed in Python with NumPy, SciPy, and Matplotlib.

4 Data

Figure 2 shows a pulse from the G-M tube observed during setup. The counts recorded in each experiment are given in the following subsections.

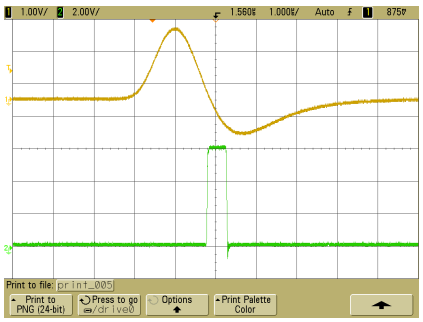


Figure 2: Oscilloscope trace of a pulse from the G-M tube at the amplifier output.

4.1 Measuring a Gaussian Distribution

The results following Experiment 2.2.1 in the lab manual are summarized in Tables 2 and 3 [2]. Table 2 lists the raw counts in the order they were taken, and Table 3 lists the number of times each count occurred.

Trial	Counts	Trial	Counts	Trial	Counts	Trial	Counts	Trial	Counts
1	19	11	26	21	29	31	31	41	27
2	30	12	34	22	36	32	21	42	20
3	36	13	32	23	40	33	33	43	29
4	28	14	29	24	32	34	26	44	38
5	30	15	35	25	28	35	38	45	24
6	26	16	28	26	23	36	25	46	32
7	46	17	28	27	30	37	31	47	33
8	34	18	37	28	27	38	33	48	29
9	33	19	37	29	26	39	40	49	40
10	36	20	23	30	25	40	34	50	25

Table 2: Raw counts recorded in each of the 50 consecutive 0.53 s intervals.

Counts	Frequency	Counts	Frequency	Counts	Frequency	Counts	Frequency
19	1	26	4	33	4	40	3
20	1	27	2	34	3	41	0
21	1	28	4	35	1	42	0
22	0	29	4	36	3	43	0
23	2	30	3	37	2	44	0
24	1	31	2	38	2	45	0
25	3	32	3	39	0	46	1

Table 3: Frequency table for the 50 measurements with a 0.53 s interval. The frequencies sum to 50.

4.2 Measuring a Poisson Distribution

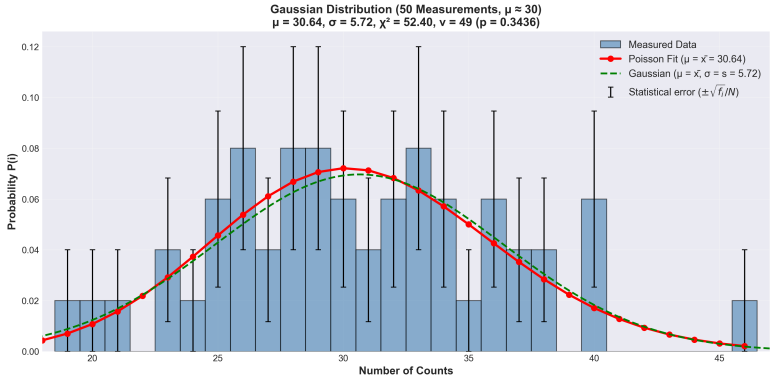
The results following Experiment 2.2.2 are summarized in Table 4. We recorded the number of counts in 1000 consecutive 0.12 s intervals. As the manual specifies, we do not include the 1000 raw values [2]. Table 4 lists the number of times each count occurred.

Counts	Frequency	Counts	Frequency
0	1	8	136
1	5	9	91
2	25	10	76
3	69	11	47
4	103	12	20
5	112	13	14
6	138	14	6
7	155	15	2

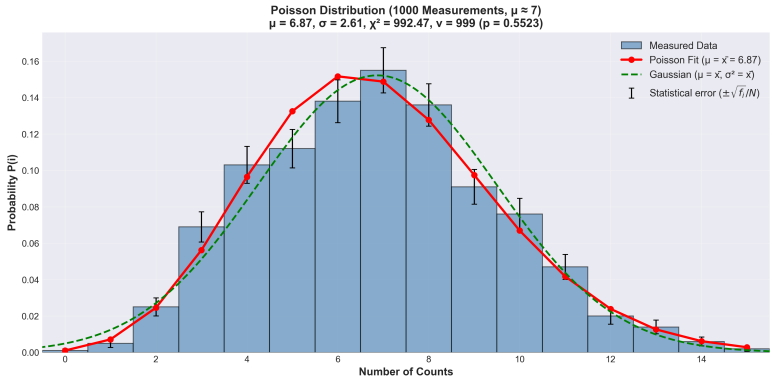
Table 4: Frequency table for the 1000 measurements with a 0.12 s interval. The frequencies sum to 1000.

5 Analysis of Data

We analyzed both experiments with the sample statistics and chi-squared test of Section 2. We took the standard deviation of a single count as $\sqrt{\bar{x}}$ and the uncertainty of the mean as $\sqrt{\bar{x}/N}$ (Knoll, Eq. 3.44) [1]. Figure 3 shows the histograms of both experiments normalized to an area of 1, with the Poisson (Eq. (3)) and Gaussian models overlaid, as the manual requires [2]. The black error bars are the statistical uncertainty $\sqrt{f_i}/N$ of each bar, where f_i is the number of measurements in the bin and N is the total number of measurements. This follows from $\sigma = \sqrt{x}$ for a count and dividing by the constant N (Knoll, Eq. 3.40) [1]. Empty bins have no bar.



(a)



(b)

Figure 3: Histograms of the measured counts, normalized for an area of 1, for (a) $N = 50$ measurements with mean $\bar{x} = 30.64$ and standard deviation $s = 5.72$, and (b) $N = 1000$ measurements with mean $\bar{x} = 6.87$. The red line with markers is the Poisson distribution at integer counts with mean $\bar{x}$, and the dashed green curve is the Gaussian distribution with mean $\bar{x}$ and standard deviation s in (a) and variance $\bar{x}$ in (b). The black error bars are the statistical uncertainty $\sqrt{f_i}/N$ of each bar.

5.1 Measurement of a Gaussian Distribution

The mean, sample variance, and standard deviation for Table 2 from Eq. (1) are

$$\bar{x} = \frac{1532}{50} = 30.64, \qquad \sigma_{\bar{x}} = \sqrt{\frac{30.64}{50}} = 0.78,$$

$$\sum_{i=1}^{50}(x_i - \bar{x})^2 = 1605.5, \qquad s^2 = \frac{1605.5}{49} = 32.77, \qquad s = 5.72.$$

The mean is about 2% of our target of 30, so the counting interval was chosen appropriately. For random counts, Eq. (4) predicts $\sigma = \sqrt{30.64} = 5.54$. The measured $s = 5.72$ is 3.4% above this value. The ratio $s^2/\bar{x} = 1.07$ is also close to the value of 1 expected for a Poisson distribution.

We then applied the chi-squared test of Eq. (7) to the departure of these data from a Poisson distribution with the same mean:

$$\chi^2 = \frac{\sum(x_i - \bar{x})^2}{\bar{x}} = \frac{1605.5}{30.64} = 52.40, \qquad \nu = 50 - 1 = 49, \qquad \frac{\chi^2}{\nu} = 1.07.$$

The probability that a random sample from a true Poisson distribution with the same mean would show a larger χ^2 is $p = 0.3436$. We calculated p from the chi-squared distribution, which is the distribution plotted in Fig. 3.11 of Knoll [1]. The probability is close to the 0.5 expected for a perfect Poisson sample and well inside Knoll's limits of 0.02 and 0.98 [1], so we found no abnormally large or small fluctuations.

In Figure 3a the Poisson and Gaussian models look very close. This is expected because the Poisson distribution approaches a Gaussian for a mean near Knoll's threshold of 25 to 30 [1]. The Gaussian is slightly broader and lower at the peak because it uses $s = 5.72$, while the Poisson distribution has the predicted $\sqrt{\bar{x}} = 5.54$. The histogram is irregular because each bar is one count out of 50, which is a height of 0.02, and the error bars are large because a bar of four counts has an uncertainty of 50%. Both curves are within two error bars of every non-empty bar, with the largest deviation about 1.5 error bars for both distributions, therefore the width and height of the curves follow the overall trend of histogram. The maximum value, 46 counts, is $2.8\sqrt{\bar{x}}$ above the mean, and the Poisson probability of a count of 46 or more is 0.57%. For 50 measurements, 0.28 such values are expected, so it is not unusual that such an outlier appeared. The deviations between the histogram and either curve are consistent with statistical fluctuation and we cannot distinguish the two models with 50 samples.

5.2 Measurement of a Poisson Distribution

The mean, sample variance, and standard deviation for the data in Table 4 are

$$\bar{x} = \frac{6867}{1000} = 6.867, \qquad \sigma_{\bar{x}} = \sqrt{\frac{6.867}{1000}} = 0.083,$$

$$\sum_{i=1}^{1000}(x_i - \bar{x})^2 = 6815.3, \qquad s^2 = \frac{6815.3}{999} = 6.822, \qquad s = 2.612.$$

The mean is about 2% of our target of 7. The predicted standard deviation is $\sqrt{6.867} = 2.620$, and the measured $s = 2.612$ differs from it by 0.3%.

For the chi-squared test,

$$\chi^2 = \frac{6815.3}{6.867} = 992.47, \qquad \nu = 1000 - 1 = 999, \qquad \frac{\chi^2}{\nu} = 0.993,$$

which gives $p = 0.5523$. For Poisson data χ^2 should be close to $\nu = N - 1$ [1], and our $\chi^2 = 992.47$ is close to 999, so this is the result we expected for Poisson-distributed counts. The probability is even closer to 0.5 than the last experiment.

In Figure 3b the two models differ more than in Figure 3a. While at first, it may seem that the Gaussian distribution fits the data better because of the means, the Poisson distribution describes the data better. First, the data is asymmetric, with a longer tail toward high counts, as the Poisson distribution predicts, whereas the Gaussian is symmetric. Second, at 0 counts the Poisson probability is $P(0) = e^{-6.867} = 0.0010$, which matches the observed one occurrence in 1000 measurements, while the Gaussian density at 0 is 0.0049, which would predict about five occurrences. The error bars show the difference more clearly. The Poisson curve is within one error bar of 12 of 16 bars and within two of all 16, with the largest deviation of 1.9 error bars at 5 counts. This aligns with random fluctuation rather than a systematic trend. The Gaussian curve is within two error bars of only 13 of 16 bars. It misses at 0, 1, and 3 counts by 3.9, 3.3, and 2.1. The mean of 6.87 is far below Knoll's threshold for the Gaussian approximation (30), which means Poisson distribution should be the appropriate model.

5.3 Comparison of the Two Experiments

Table 5 compares the results of the two experiments. Both data sets are consistent with Poisson statistics, since s agrees with $\sqrt{\bar{x}}$ and the chi-squared probability is near 0.5 in each case. At a mean near 30 the Gaussian and Poisson models cannot be distinguished, while at a mean near 7 the Poisson distribution is the better model, aligning with the guideline of using the Poisson distribution for counts below about 30 [1].

Quantity	$N = 50$	$N = 1000$
Counting interval (s)	0.53	0.12
Mean $\bar{x}$ (counts)	30.64 ± 0.78	6.867 ± 0.083
Sample variance s^2	32.77	6.822
Sample standard deviation s	5.72	2.612
Predicted $\sigma = \sqrt{\bar{x}}$	5.54	2.620
$s^2/\bar{x}$	1.07	0.993
χ^2	52.40	992.47
Degrees of freedom ν	49	999
χ^2/ν	1.07	0.993
p	0.3436	0.5523

Table 5: Summary statistics and chi-squared results for both experiments. The uncertainty on the mean is $\sqrt{\bar{x}/N}$.

We note that the chi-squared test of Eq. (7) only checks that the variance equals the mean. Both the Gaussian and Poisson distributions have $\sigma^2 = \bar{x}$, so the test cannot distinguish a Gaussian from a Poisson distribution with the same mean. It is better to compare two models from the shape of the distributions in Figure 3b.

6 Conclusions

This laboratory demonstrated that counts from a G-M tube and thorium source follow the Gaussian and Poisson statistical models. We recorded 50 counts with a mean of 30.64 and 1000 counts with a mean of 6.87. In both cases the sample standard deviation agreed with the Poisson prediction $\sqrt{\bar{x}}$ (5.72 compared with 5.54, and 2.61 compared with 2.62), and the chi-squared

tests gave $\chi^2/\nu = 1.07$ ($p = 0.344$) and 0.993 ($p = 0.552$). Both probabilities are close to the 0.5 expected for a perfect Poisson sample and inside Knoll's limits of 0.02 and 0.98. We therefore found no abnormally large or small fluctuations, and the uncertainty of a single count can be taken as $\sqrt{N}$.

At a mean near 30 the Gaussian and Poisson models could not be easily distinguished with 50 measurements (Figure 3a). At a mean near 7 the Poisson distribution was the better model (Figure 3b). The chi-squared test checks only the variance, so it cannot distinguish the two models itself. The main limitation of the experiment is the small number of samples in the first data set, which gives an irregular histogram and a much less sensitive chi-squared test. Future work for this laboratory would be to record more than 50 counts at a mean near 30. We could also repeat the measurements at several count rates and source distances to analyze for dead-time effects.

References

[1] G. F. Knoll, *Radiation Detection and Measurement*. Hoboken, NJ: John Wiley & Sons, Inc., fourth ed., 2010.

[2] University of Michigan, Department of Nuclear Engineering and Radiological Sciences, *NERS 515 Nuclear Measurements Laboratory: Laboratory 3*, 2026.